Part E: Development Tools & Process Setup — DebateSpeak AI

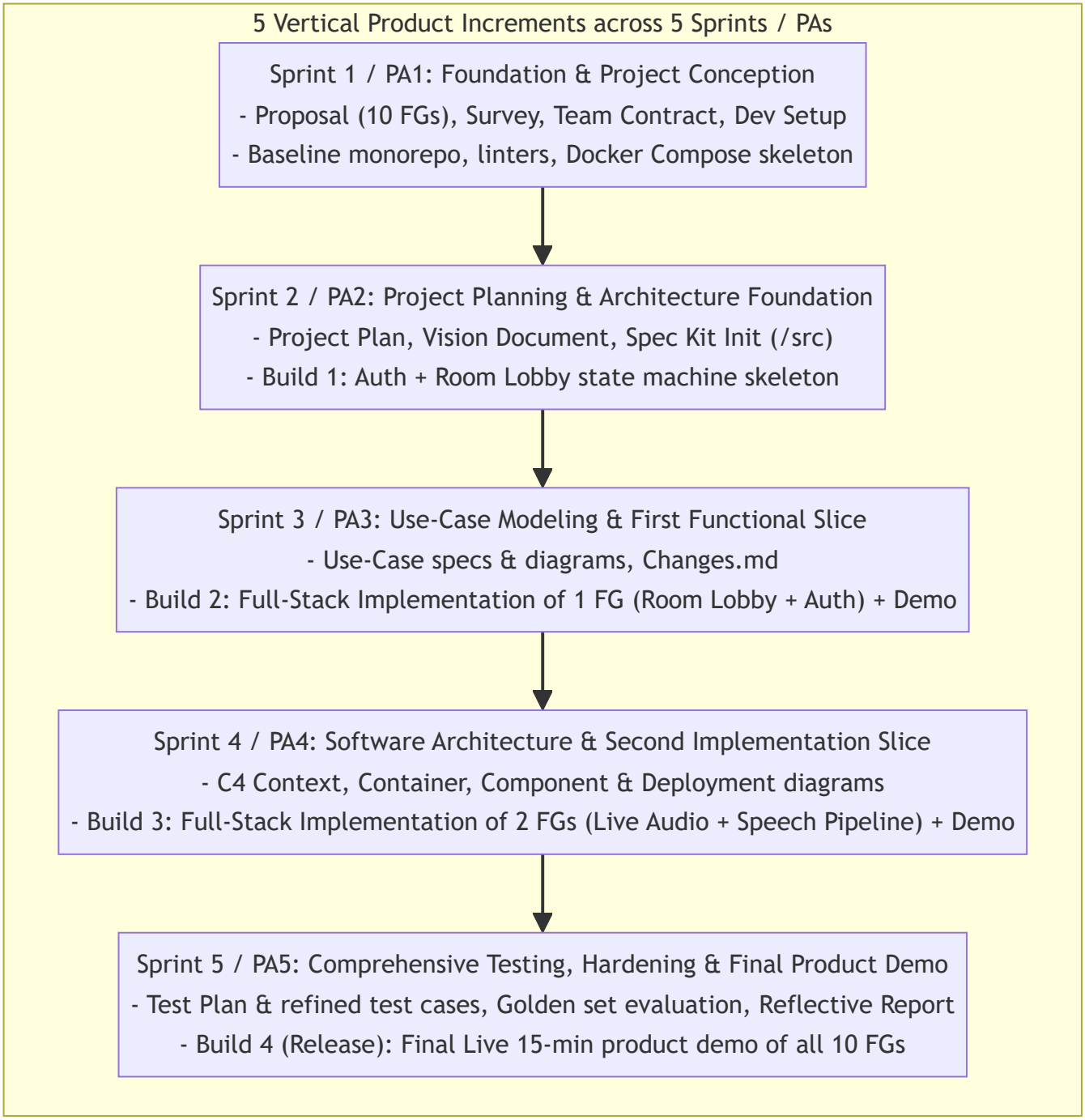
Field	Description
Course	CS300 / CSC13002 — Introduction to Software Engineering
Document	**Part E: Scrum Process, Engineering Tooling Setup & Multi-Sprint WBS**
Score Allocation	Part E (5 Points)

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1. Scrum Methodology & Incremental Development Strategy (5 Sprints = 5 PAs)

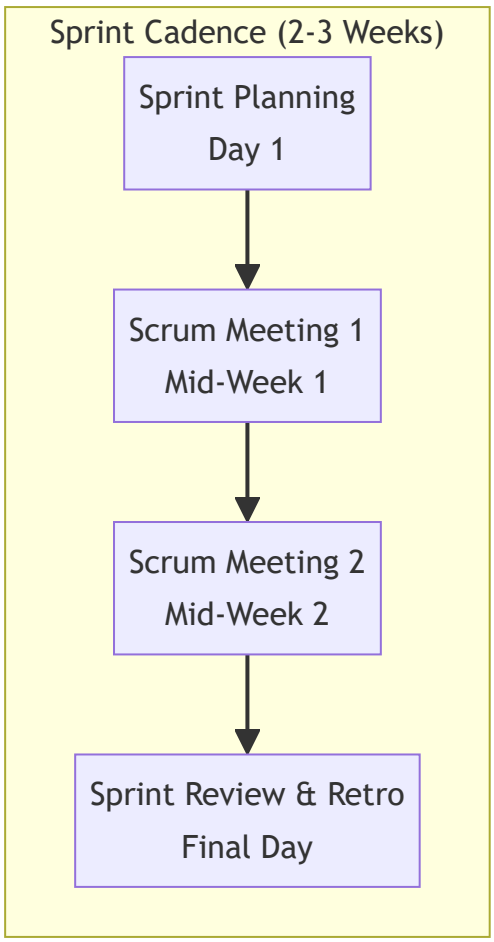
Performed by: Nguyễn Đình Thiên Lộc (24125093) | Reviewed by: Nguyễn Bảo Minh Triết (24125047) | Edited by: Nguyễn Hồng Tấn Tài (24125078)

Our team adheres strictly to the **Agile Scrum framework combined with an Incremental Development model** structured across **5 Sprints**, corresponding directly to the 5 course Project Assignments (PA1–PA5):



Incremental Development Principles

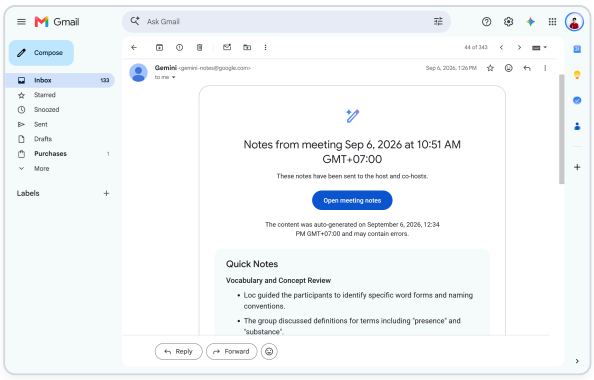
1. **Vertical Slices Over Horizontal Layers:** Each increment integrates frontend UI, backend logic, and database persistence so that the increment is immediately testable by end users.
2. **Continuous Integration & Regression Guarding:** Code from each increment is merged into `develop` only after automated tests pass. Existing working functionality is never broken to support upcoming increments.
3. **Spec Kit & Traceability:** Starting from PA2 initialization, Spec Kit artifacts guide feature implementations with strict traceability from use cases to tests.



- **Sprint Planning Meeting (Day 1 of Sprint):** Review course PA requirements, decompose backlog into Jira user stories, estimate story points, and commit to the Sprint Backlog.
- **Scrum Check-ins (2 Meetings per Sprint):** Held mid-week to evaluate progress against the sprint burndown chart, identify emerging technical blockers, and coordinate API contracts.
- **Sprint Review & Retrospective (Final Day of Sprint):** Demonstrates working software against acceptance criteria, conducts an honest team retrospective, and compiles PA submission artifacts.

Team Meeting Evidence & Automated AI Minutes

As part of our Scrum cadence, team meetings are conducted synchronously via Google Meet with automated AI note-taking and minutes extraction (Gemini in Google Meet), generating structured meeting artifacts distributed directly to team members:

Live Scrum Meeting Session	Automated Gemini Meeting Minutes
	
Figure 1.1: Live Scrum check-in meeting on Google Meet with Gemini AI note-taking activated.	*Figure 1.2: Automated meeting notes, concept review, and action items distributed via email.*

2. Jira Task Management Standards

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The team utilizes **Jira** for end-to-end task tracking following strict course guidelines:

1. **Complete Project Logging:** All activities—including coding, writing documentation reports, research spikes, self-training, and setting up CI pipelines—are logged as formal Jira issues.
2. **Single Assignee Invariant:** Every Jira issue is assigned to **exactly one team member** (no shared tickets). Collaborative tasks are decomposed into distinct paired sub-tasks.
3. **Strict Date Progression:**
    - Every issue records an explicit **Creation Date**, **Assignment Date**, and **Completion Date**.
    - Tasks are created and assigned *during Sprint Planning before work begins*.
    - Tasks are completed individually as work finishes throughout the sprint—**no batch completions at the end of sprints**.
4. **Task Estimates:** Every task has an initial Story Point estimate (1, 2, 3, 5, 8) and an assigned priority (`BLOCKER`, `CRITICAL`, `MAJOR`, `MINOR`).

3. Git Workflow & Version Control Invariants

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- **Branching Strategy (Git Flow):**
    - `main`: Production-ready, stable releases (tagged `v1.0-pa1`, `v2.0-pa2`, etc.). Protected branch; direct pushes prohibited.
    - `develop`: Shared integration branch for active sprint development.

    - `feature/<jira-key>-<short-description>`: Individual feature branches branched off `develop`.
    - `bugfix/<jira-key>-<short-description>`: Defect fixes branched off `develop`.

- **Sprint 1 Centralized Documentation Custodian Role:**
    - During Sprint 1 (conception & foundation phase), all five team members collaborated in drafting, review, and editing of project deliverables (documented verbatim via per-section headers).
    - To prevent Git merge conflicts during rapid simultaneous drafting of Markdown files and ensure uniform LaTeX compilation via `latexmk`, **Nguyễn Đình Thiên Lộc served as Documentation Custodian & Git Release Manager**, centralizing integration, running linters, and packaging assignment archives.
    - Starting from Sprint 2 (PA2), each team member will independently branch, commit, and open PRs for their assigned vertical slice implementation tasks.

- **Pull Request Rules:**
    - Every PR requires passing automated CI checks (linting, tests, build).
    - Every PR requires at least **one approved code review** from a designated peer reviewer before merging.

- **Environment Agnostic & Secret Invariants:**
    - **Zero Hardcoded Absolute Paths:** Dynamic relative paths or environment configurations only.
    - **Zero Secret Leaks:** `.env` is strictly git-ignored; only sanitized `.env.example` is committed.

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4. Spec-Driven Development & Spec Kit Integration

Performed by: Trần Lê Anh Tuấn (24125107) | Reviewed by: Nguyễn Đình Thiên Lộc (24125093) | Edited by: Nguyễn Anh Khoa (24125058)

- In accordance with course guidelines, the team adopts **Specification-Driven Development** starting from PA1 and preparing for **Spec Kit** initialization in PA2 (in `/src`).
- Before code implementation begins on any feature:
    1. A formal specification (data models, Pydantic schemas, API route contracts, and edge cases) is written and reviewed in `docs/architecture/SOFTWARE_ARCHITECTURE_SPEC.md`.
    2. Test cases and acceptance criteria are authored in `docs/testing/TESTING_PLAN.md`.
    3. AI coding assistants (GitHub Copilot / Cursor) are constrained using these specifications as prompt context, ensuring code generation directly adheres to architecture invariants.

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5. AI Coding Tooling & Educational Accounts

Performed by: Nguyễn Đình Thiên Lộc (24125093) | Reviewed by: Nguyễn Anh Khoa (24125058) | Edited by: Trần Lê Anh Tuấn (24125107)

All 5 team members have registered and verified educational developer accounts:

- **GitHub Copilot for Students:** Activated via the GitHub Student Developer Pack on university email domains (`@hcmus.edu.vn`). Used for inline autocompletion and unit test scaffolding.
- **Codex/ Antigravity:** Configured with local project context (`AGENTS.md`, `.cursorrules`) to assist with full-stack refactoring, FastAPI schema validation, and React component styling under strict quote-grounding constraints.

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6. Sprint 1 Work Breakdown Structure (PA1 Tasks)

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Task Key	Task Summary	Assignee	Issue Type	Story Points	Target Completion
`DSP-1`	Author Project Proposal & 10 Functional Groups	Nguyễn Bảo Minh Triết (24125047)	Documentation	5 pts	Week 1, Day 4
`DSP-2`	Conduct Existing App Survey (ArguFight, Khaos Live, ELSA Speak)	Nguyễn Hồng Tấn Tài (24125078)	Research/Doc	5 pts	Week 1, Day 6
`DSP-3`	Draft Team Contract & Accountability Guidelines	Nguyễn Anh Khoa (24125058)	Documentation	3 pts	Week 2, Day 2
`DSP-4`	Author Software Architecture & System Spec Document	Trần Lê Anh Tuấn (24125107)	Technical Spec	8 pts	Week 2, Day 4
`DSP-5`	Initialize Git Monorepo, Directory Hierarchy & CI Linters	Nguyễn Đình Thiên Lộc (24125093)	DevOps/Setup	3 pts	Week 1, Day 3
`DSP-6`	Setup Docker Compose (PostgreSQL, Redis, LiveKit dev)	Nguyễn Đình Thiên Lộc (24125093)	DevOps/Setup	3 pts	Week 2, Day 3
`DSP-7`	Verify Educational AI Coding Accounts (Copilot/Cursor)	All	Administrative	1 pt	Week 1, Day 2
`DSP-8`	Setup `latexmk` pipeline & compile academic PA1 report	Nguyễn Đình Thiên Lộc (24125093)	Management	2 pts	Week 2, Day 5
`DSP-9`	Compile PA1 Markdown Package & Export Submission PDFs	Nguyễn Bảo Minh Triết (24125047)	Management	2 pts	Week 2, Day 6